

Assignment 2 (Lab Report: BLASTN)

Basic Bioinformatics (BM2023)

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Part A

Assignment Objective

In a pandemic the infective material is tested for DNA and further sequenced. The following nucleotide sequence was found. Using BLASTN, identify the DNA segment from which the sequence shows alignment and describe the gene function.

Materials and Methods

Materials

- Query nucleotide sequence (850 bp DNA fragment obtained from infective material)
- NCBI BLASTN online tool
- NCBI nucleotide (nt) database
- Computer system with internet access

Methods

1. The provided DNA sequence was submitted to the NCBI BLASTN web interface.
2. The nucleotide database (nt) was selected for alignment.
3. The megablast algorithm was used for detecting highly similar sequences.
4. Default alignment parameters were applied.
5. Significant matches were evaluated based on query coverage, percent identity, and E-value.
6. The top hit with the highest query coverage and lowest E-value was selected for interpretation.

Results

Top BLASTN Match

- **Gene:** Genome assembly (chromosome 5)
- **Organism:** *Sitta europaea*
- **Query Coverage:** 96%
- **Percent Identity:** 92.81%
- **E-value:** 3×10^{-165}

The alignment shows that approximately 96% of the query sequence matches the *Sitta europaea* genome assembly (chromosome 5) with very high identity and an extremely low E-value, indicating strong statistical significance.

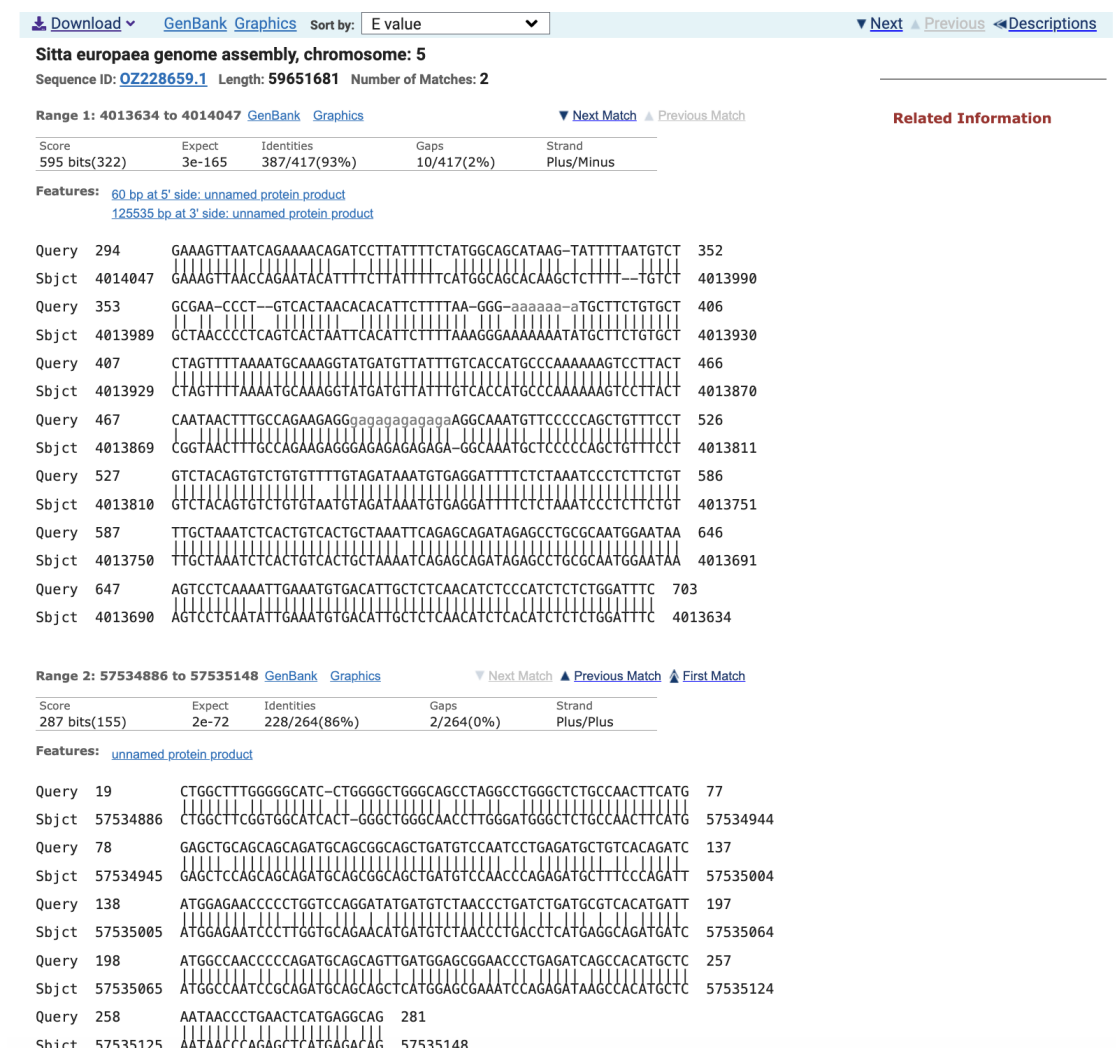


Figure 1: BLASTN alignment of query sequence with *Sitta europaea* chromosome 5

Gene Function

The top BLASTN alignment corresponds to chromosome 5 of *Sitta europaea* (Eurasian nuthatch). The alignment shows 96% query coverage with 92.81% percent identity and an E-value of 3×10^{-165} , indicating a highly significant match.

Since the hit corresponds to a genome assembly region rather than a specifically annotated gene, the sequence likely represents a conserved genomic segment within chromosome 5. Such regions may include:

- Protein-coding genes (coding sequences)
- Regulatory elements controlling gene expression
- Intronic regions

- Conserved structural DNA regions

The extremely low E-value confirms that the alignment is not due to random chance, and the high percent identity suggests strong evolutionary conservation. Therefore, the query sequence is identified as part of a conserved genomic region of *Sitta europaea*.

Conclusion

Based on BLASTN analysis, the provided nucleotide sequence shows a highly significant alignment with chromosome 5 of *Sitta europaea*. The alignment demonstrates 96% query coverage, 92.81% percent identity, and an E-value of 3×10^{-165} , confirming that the match is statistically robust.

Since the alignment corresponds to a genomic assembly region rather than a specifically annotated gene, the sequence is identified as part of a conserved chromosomal segment. The high percent identity suggests strong evolutionary conservation, indicating that the sequence likely represents a functional genomic region within chromosome 5 of *Sitta europaea*.

Thus, the infective material DNA fragment is concluded to originate from a conserved genomic region of *Sitta europaea*.

Part B

The overall base composition of the *E. coli* genome is such that $A = T = 24.6\%$ and $C = G = 25.4\%$.

Assuming a random DNA sequence of length 4,639,221 bp with these base proportions and independent nucleotide distribution, the expected number of occurrences of the sequence CTAG is calculated as follows.

Solution

Step 1: Base Probabilities

$$P(A) = 0.246$$

$$P(T) = 0.246$$

$$P(C) = 0.254$$

$$P(G) = 0.254$$

Step 2: Probability of CTAG

$$P(\text{CTAG}) = P(C) \cdot P(T) \cdot P(A) \cdot P(G)$$

$$= (0.254)^2(0.246)^2$$

$$= 0.003903$$

Step 3: Number of Possible Positions

$$N - k + 1 = 4,639,221 - 4 + 1 = 4,639,218$$

Step 4: Expected Number

$$E[\text{CTAG}] = 4,639,218 \times 0.003903$$

$$\approx 18,106$$

Expected occurrences of CTAG $\approx 1.81 \times 10^4$
